

Literature-Implied Partial Answer to a Block-Minimal/Meager-Isomorphism Problem

Run `fa_banach_001`, lane 17

June 19, 2026

Status

literature_implied_answer (partial local-block-minimal variant). This is not a full solution of the source problem.

Source Problem

The source paper is V. Ferenczi and G. Godefroy, *Tightness of Banach spaces and Baire category*, arXiv:1111.6444. At the end of Section 4, after Example T concerning T^* , the authors ask:

Find a space X which embeds as a block-subspace of all its block-subspaces, but such that Is is meager.

Here Is denotes the isomorphism relation on the Polish space $b(X)$ of block-subspaces.

Supporting Result

The supporting paper is V. Ferenczi and C. Rosendal, *Banach spaces without minimal subspaces*, arXiv:0711.1350. In Section 6, Example `OdellSchlumprecht`, they prove:

There exists a reflexive, tight, locally block minimal Banach space.

Identification

The source paper proves in Proposition 3.3 that if X is tight, then the embeddability relation Emb is meager in $b(X)^2$. Since

$$\text{Is} \subseteq \text{Emb},$$

the isomorphism relation Is is also meager for every tight space. Combining this with arXiv:0711.1350 gives a reflexive Banach space whose basis is locally block minimal and whose block-subspace isomorphism relation is meager.

Thus the natural local-block-minimal weakening of the source problem has a positive answer in the literature.

Limitations

This is an agent-identified implication rather than an explicit answer by the supporting authors to arXiv:1111.6444. It does not give a space which embeds as a block-subspace of all its block-subspaces. Local block minimality is a finite-dimensional block finite-representability condition, weaker than block-minimality. The exact source problem therefore remains open from this packet's point of view.

Search Evidence

The local run indexes were searched for arXiv:1111.6444, the source title, **block-minimal**, **block minimal**, **locally block minimal**, **Is is meager**, and the exact problem sentence. No existing packet for this implication was found. Bounded web searches for the exact block-minimal/meager problem and close variants did not reveal a later exact solution.

References

- [1] V. Ferenczi and G. Godefroy, *Tightness of Banach spaces and Baire category*, arXiv:1111.6444.
- [2] V. Ferenczi and C. Rosendal, *Banach spaces without minimal subspaces*, arXiv:0711.1350.